Heart Disease Prediction Using Supervised Machine Learning

Dossiya Dakou
School of Sustainable Engineering and the Built Environment
Arizona State University
Tempe, AZ, USA
Email: ddakou@asu.edu

***Abstract*—This paper analyzes a heart-disease dataset in the context of developing a predictive model for clinical decision support. The dataset contains 270 patient records with 13 clinical and demographic variables and a binary outcome variable indicating heart disease status. A statistical study was conducted using correlation, covariance, and visualization techniques to identify feature relationships and predictive structure. Multiple supervised learning algorithms were evaluated. Logistic Regression achieved the highest test accuracy (0.8519) with strong generalization performance. The results indicate that prediction relies on multivariate relationships rather than a single dominant feature.**

I. Introduction

This study evaluates a heart disease dataset using statistical analysis and supervised machine learning to support clinical decision-making. The goal is to understand feature relationships and assess predictive performance.

II. Dataset and Objective

The objective is to estimate:

$$y = f(X) \tag{1}$$

where $X = (x_1, x_2, ..., x_{13})$ represents patient features and $y \in \{1, 2\}$ represents disease status.

All features are retained without dimensionality reduction.

III. Statistical Analysis

A. Correlation Analysis

The Pearson correlation coefficient is defined as:

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \tag{2}$$

Absolute correlation values $|r|$ are used to measure association strength independent of sign.

The strongest predictor–target relationships are:

$$|r(\text{thal}, y)| = 0.525, \quad |r(\text{nmvcf}, y)| = 0.455 \tag{3}$$

A second group of moderately predictive variables satisfies:

$$|r| \approx 0.418 \text{ for } \{\text{eia, mhr, opst, cpt}\} \tag{4}$$

No feature satisfies:

$$|r(x_i, y)| \approx 1 \tag{5}$$

indicating that prediction requires multivariate relationships.

The strongest predictor–predictor correlation is:

$$|r(\text{opst}, \text{dests})| = 0.610 \tag{6}$$

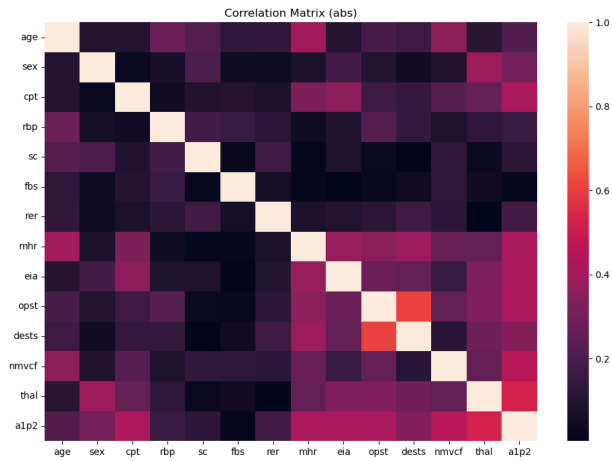

Fig. 1. Absolute correlation coefficient matrix $|r|$ for all variables.

B. Covariance Analysis

Covariance is defined as:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \tag{7}$$

Covariance is scale-dependent:

$$\text{Cov}(aX, bY) = ab \cdot \text{Cov}(X, Y) \tag{8}$$

Large covariance values reflect measurement scale rather than predictive importance.

C. Pair Plot Analysis

Observations:

- Continuous variables appear unimodal to moderately skewed
- Categorical variables produce banded patterns
- Strong class overlap is observed

This implies:

$$P(y = 1|x_i, x_j) \approx P(y = 2|x_i, x_j) \tag{9}$$

and therefore:

$$\text{Low separability in } \mathbb{R}^2 \tag{10}$$

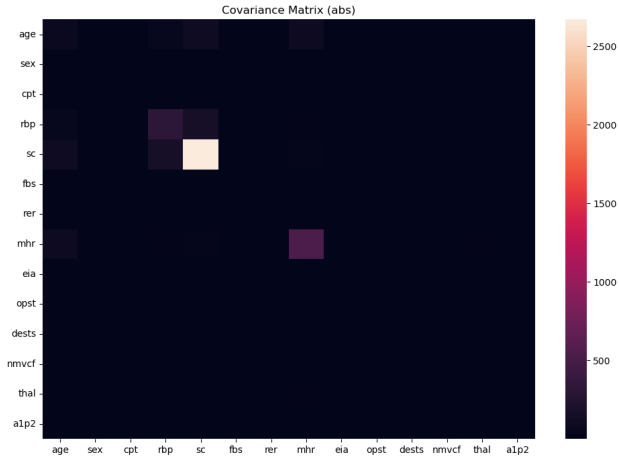

Fig. 2. Absolute covariance matrix (scale-dependent).

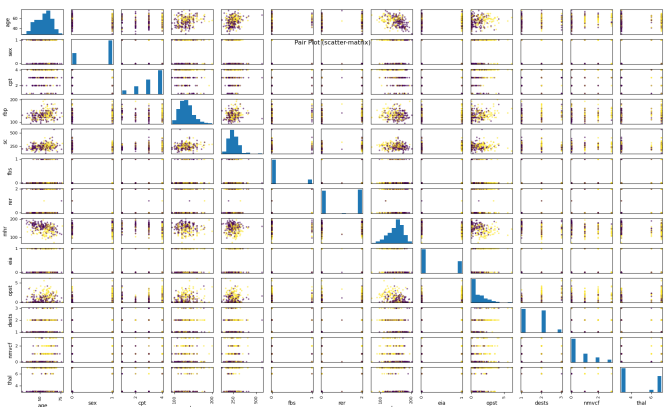

Fig. 3. Pair plot (scatter-matrix) of predictors colored by class.

IV. Modeling

Logistic Regression models probability as:

$$P(y = 1|X) = \frac{1}{1 + e^{-\beta^T X}} \tag{11}$$

Accuracy is defined as:

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(y_i = \hat{y}_i) \tag{12}$$

V. Results

VI. Discussion

The dataset satisfies:

$$\max_i |r(x_i, y)| \approx 0.525 < 0.7 \tag{13}$$

indicating no dominant predictor.
Prediction requires:

$$f(X) = f(x_1, x_2, ..., x_{13}) \tag{14}$$

The strongest feature dependency:

$$|r(\text{opst}, \text{dests})| = 0.610 \tag{15}$$

indicates moderate redundancy.

TABLE I
Model performance comparison

Method	Accuracy	Combined	Gap
Logistic Regression	0.8519	0.8667	0.0148
KNN	0.8395	0.8593	0.0198
Perceptron	0.8148	0.8370	0.0222
Random Forest	0.8148	0.9444	0.1296
SVM	0.7901	0.8926	0.1025
Decision Tree	0.7284	0.8889	0.1605

VII. Conclusion

Logistic Regression is recommended because:

- highest test accuracy
- small generalization gap
- interpretability

The model achieves:

$$\text{Gap} = 0.0148 \tag{16}$$

indicating low overfitting and strong generalization.

Future work includes cross-validation, sensitivity analysis, and nonlinear modeling.